University of Wisconsin
SCHOOL OF MEDICINE
AND PUBLIC HEALTH

PH-STAT
(Topological Inference for Persistent Homology)

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What is PH-STAT

PH-STAT is a stand-alone MATLAB toolbox for topology computation and inference for various biomedical data under development.

Downloading PH-STAT

https://github.com/laplcebeltrami/PH-STAT

It ha … hlx.

hierarchical_cluster.m	Add files via upload	3 years ago
kmeans_cluster.m	Add files via upload	3 years ago
online_pvalues.m	Add files via upload	3 years ago
plot_distribution.m	Add files via upload	10 months ago
randcorr.m	Add files via upload	3 years ago

README

PH-STAT (Statistical Inference on Persistent Homology)

Standalone MATLAB Toolbox for Statistical Inference on Persistent Homology

Currently under development but we made some of codes available for test run and debug purpose. Please provide feedback and corrections. Simply load MATLAB live editor file SCRIPT.mlx. The detailed manual is available here (last updated on Feb, 2025) or as arXiv at arXiv:2304.05912. If you are using or modifying these tools, please reference arXiv.

The package can perform the following topological tasks: Rips filtration, graph fltration, topological clustering, topological embedding. About 80% of functionalities are not implmented yet. The codes are constnatly updated and debugged.

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Last update October 22, 2025

Manual for PH-STAT

PH-STAT: Statistical Inference on Persistent Homology

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August 1, 2025

Abstract

We introduce PH-STAT, a comprehensive MATLAB toolbox designed for performing a wide range of statistical inferences and learning tasks on persistent homology, primarily for network and graph data, with an emphasis on network analysis. Persistent homology is a prominent tool in topological data analysis (TDA) that captures the underlying topological features of complex data sets. The toolbox aims to provide users with an accessible and user-friendly interface for analyzing and interpreting topological data. The Matlab package is distributed in https://github.com/laplcebeltrami/PH-STAT.

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https://arxiv.org/abs/2304.059

Doing project with PH-STAT

Most projects can be done with by modifying the core codes in PH-STAT.

Hodge Laplacian (Graph Laplacian part) is not yet implemented. This allows the spectral modeling of topology. But Hodge Laplacian code can be found in https://github.com/laplcebeltrami/hodge

These two code bases are all you need to do topology related projects.

PH-STAT

- Allows topological averaging
- Spectral expansion of topology
- Topological distance based inference and clustering
- Provide various sample biomedical data to load into package and play with – not yet implemented